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Modelling the impact of business risk on the competitiveness of purpose-grown biomass of annual and perennial crops

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ABSTRACT

Perennial energy crops compete with conventional annual crops in land-use. The production price of biomass from perennial energy crops is thus derived on the basis of the opportunity cost principle. This means that in deciding whether to grow perennial energy crops, the farmer will demand a price for biomass that will at least provide him with the same economic effect over the lifetime of the plantation in the form of the present value of net cash flows over the lifetime of the energy crop plantation. The level of discounts used to discount the cash flows associated with (annual) conventional crops and perennial energy crops plays an important role in modelling the biomass production price. The business risk associated with the two types of crops is different, with generally higher risk associated with perennial energy crops. This study focuses on developing a methodology to incorporate the risk associated with growing conventional and energy crops and to derive the discount rate for both crop groups. The methodology is demonstrated using the example of the Czech Republic and the price level in 2021. The analysis using Czech data led to a discount rate of 8.5 % for conventional crops and 11.5 % for perennial energy crops. The increase in the discount value for perennial energy crops leads to an increase in the expected biomass sales price by a producer. The sensitivity of the biomass production price level to an increase in the discount rate is lower for *Miscanthus* plantations compared to short rotation coppice.

1. Introduction

The share of renewable energy sources (RES) in the EU's gross final energy consumption reached 21.8 % in 2021, doubling over the period 2004–2021 [1]. Although the relative share of biomass in RES energy consumption is decreasing due to the rapid development of other types of RES, its share remains high, reaching about 50 % of gross available energy from RES in 2020 [2]. The most important type of biomass is woody and non-woody biomass, which reaches about 75 % of the total biomass [3]. The share of biomass derived from agricultural land is increasing and reaches approximately 20 % in 2019 [4].

The use of residual biomass from conventional crops contributes significantly to the potential of biomass for energy purposes. Residual biomass from conventional crops has a wide range of uses from direct combustion (e.g. straw burning in heating plants, transformation to solid, liquid or gaseous biofuels). For example [5], presents a bio-economic model for optimizing the use of residual biomass from artichoke cultivation for processing in biogas plants. Based on the modelling results, it then demonstrates that the use of locally available residual biomass can lead to a reduction in the requirement for land take for targeted energy crops in regions where this residual biomass has a direct use. The spatial distribution of residual biomass thus plays an important role in terms of its potential for use and inclusion in the useable biomass potential.

In general, however, it can be stated that the sources of residual and waste biomass from the wood processing industry, from forest residues and fuelwood, and from agriculture are already substantially allocated for use and the growth potential of biomass from these sources is limited [6]. An option for increasing the potential of biomass as a renewable energy source is targeted cultivation on agricultural land while maintaining sustainability criteria. Thus, sustainability and maximizing the positive ecological effects associated with plantations of perennial energy crops like fast-growing trees, *Miscanthus*, Reed Canary grass or Schavnat, would become an important consideration in the use of agricultural land for energy crops [7].

Increasing energy crops grown on agricultural land brings up a number of issues, such as the possibilities for the development of RES use in a given country, the strategy for the development of agricultural land-use in a given country, changes in land management practices (e.g.

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NOMENCLATURE

Abbreviations

CAPM	Capital Assets Price Model
CC	Conventional crop
CF	Cash flow
EC	Energy crop
EU	European Union
EUR	Euro
GIS	Geographic information system
JČ	Jihočeský region
JM	Jihomoravský region
KV	Karlovarský region
KR	Královehradecký region
LI	Liberecký region
LPIS	Land Parcel Identification System
MO	Moravskoslezský region
MSCU	Main soil and climate unit
NPV	Net present value
NUTS3	Territorial statistical unit (region)
OL	Olomoucký region
PA	Pardubický region
PL	Plzeňský region
RES	Renewable energy sources
S1	Scenario of discount rates (8,5 % for CC and 8,5 % for EC)
S2	Scenario of discount rates (8,5 % for CC and 11,5 % for EC)
S3	Scenario of discount rates (8,5 % for CC and 14,5 % for EC)
SAPS	Single area payment scheme
SRC	Short rotation coppice
ST	Středočeský region
ÚS	Ústecký region
VUKOZ	Silva Tarouca Research Institute for Landscape and Ornamental Gardening
VY	Vysočina region
WACC	Weighted avarage cost of capital
ZL	Zlínský region

Notations/Symbols

k	Serial year in the lifetime of a perennial plantation
$C_{alt,1}$	Production price of biomass from the perennial energy crop in the first year of plantation lifetime
T_{lf}	Lifetime of plantation of perennial energy crop
$QBio_k$	Biomass yield of a given perennial energy crop in year k on a given land plot
$SBio_k$	Subsidy for the energy crop in year k
$EBio_k$	All one-off and operating expenses of the energy crop plantation incurred in year k, including finance costs and taxes
$r_{n,Bio}$	Nominal discount for discounting cash flows from plantations of perennial energy crop
$Qcon_{q,k}$	Yield of the q-th conventional crop in the crop rotation on the land plot in year k
$Ccon_{q,k}$	Price of the q-th conventional crop in the year k

Units

EUR	Euro
GJ	Gigajoule
PJ	Petajoule
t	Tonne

requirements to reduce the use of artificial fertilizers), and changes in subsidy policy in agriculture. The economic competitiveness of (perennial) energy crop cultivation versus conventional agricultural production is also an important consideration for the development of energy crop cultivation on agricultural land.

The strategy for the further development of energy crop cultivation on agricultural land and the estimation of its potential must consider the economic competitiveness of biomass and farmers' decision-making based on economic advantage. This comparison should cover the whole life cycle of a perennial energy crop plantation, while also considering the different level of risk associated with growing (annual) conventional and perennial energy crops. Trying to determine the potential of biomass production without considering the competitiveness aspect may lead to significant deviations from the actual biomass results. This deviation increases the risk of failure to fulfil the original intentions of biomass use (e.g. in the State energy policy) [8]. Farmers' decisions about the types of crops to be grown and their business strategy are influenced by a number of factors other than economic efficiency, such as the overall business context of the farm, the orientation towards short- or long-term goals, including an emphasis on sustainable land-use considerations. Decisions of farmers are strongly influenced also by financial subsidy schemes of Common Agricultural Policy (CAP 2023+) which in lasts years more and more supports environmental functions of agriculture. Among new approaches supported by CAP are also multi-cropping systems like agroforestry system and different eco-schemes (SRC). Private business schemes [9] currently support carbon and regenerative farming [10] which are possible to combine with energy crops especially trees.

Energy crops compete with conventional crops in land-use, and this leads to competition between food production and biomass production for energy purposes. For example [11], emphasises that the use of so-called marginal lands can be an effective solution to the conflict between land-use for energy, food production and the environment. However, even here the question of the economic decision between conventional crops and energy crops remains open. Another factor that plays a role here is that perennial energy crops provide important ecosystem services by breaking up large swathes of conventional crops. The benefits are then, for example, reducing the risk of soil erosion, increasing biodiversity, increasing the soil's capacity to absorb rainfall, etc. [12] Another major advantage of perennial energy crops over conventional annual crops is their higher capacity to sequester carbon in the soil [13]. As long as perennial energy crops are used as an ecosystem tool, the competition between conventional and perennial crops must again be addressed and the level of risk associated with both types of crops correctly determined.

1.1. Approaches to pricing biomass from perennial energy crops

The economic competitiveness of biomass from perennial energy crops is one of the key factors influencing not only the estimation of their total potential for the given area in the future, but also determines the priorities in the development of bioenergy (types of energy crops, methods of transformation of raw biomass into solid, liquid or gaseous biofuels). Two basic methodological approaches can be found in the literature for assessing the economic efficiency of biomass production from perennial energy crops and for determining production costs and production prices, respectively.

The first methodological approach is based on the calculation of the cost of biomass production over the entire life cycle of an energy crop plantation, from its establishment to its disposal at the end of its useful life. Two basic approaches are used to calculate the costs - a static approach without respecting the time value of money and the evolution of biomass input prices, and an approach based on modelling nominal cash flows over the lifetime of the plantation and respecting the time value of money. An example of the static approach is the methodological approach used for cost analysis of biomass production from perennial energy crops in the Malopolska region [14]. Here, the authors analyse biomass production costs by calculating the average annual cost per ton of biomass. The one-off costs associated with the establishment of plantations, or their disposal at the end of their lifetime, are spread over the years of the plantation's lifetime using depreciation determined as a ratio of one-off costs and lifetime. Operating costs include the costs of all agro-operations such as harvesting, fertilisation, removal of biomass from the field, etc., as well as costs associated with land rent, taxes and other financial costs. This simplified approach to determining the costs of biomass production leads to an overestimation of the weight of one-off costs associated with the establishment and liquidation of the plantation, and does not allow for the expected price evolution of the various inputs needed for biomass production.

A more comprehensive approach to biomass costing, considering the specific conditions of the business environment, including the time value of money, is presented, for example, in Ref. [15]. Here, the authors use a methodology based on modelling nominal after-tax cash flows over the lifetime of the plantation and determining the so-called minimum production price. This is the price that provides the investor in biomass production from plantations of perennial energy crops with the required return on capital employed. For practical reasons (see e.g. Refs. [8,15]), it is appropriate to express biomass production in GJ or GJ/ha while respecting the calorific value corresponding to the harvest moisture.

However, this is a value at the "border" of the field, or after the biomass has been transported to a nearby handling site, and does not include other logistics cost. [16], based on an analysis of the entire fuel cycle, points to a significant price increase for a conventional customer taking solid biofuels from such produced biomass.

The second basic methodological approach to assessing the economic efficiency of biomass production from perennial energy crops is to derive the biomass production price based on the opportunity cost principle. This is based on alternative land-use options for conventional agricultural production. For example [17], emphasises the opportunity cost principle in modelling the global supply (availability) of biomass, which reflects three basic factors, namely differences in land productivity between regions, differences in input prices and, at the same time, agricultural commodity prices. [18] stresses the need to use a fair comparative basis when comparing the economics of perennial energy crops with those of annual conventional crops. Biomass from plantations of perennial energy crops is characterized using the so-called "annual equivalent value" and compares it with the annual gross margin of conventional crops. An approach to pricing biomass from perennial energy crops based on competition for land between conventional and energy crops is also presented in Ref. [19]. It incorporates a behavioural aspect into the economic-based decision making, where a farmer's decision to grow perennial energy crops is influenced by his past experience or the experience of other farmers operating in his locality. Through this experience, the risk associated with growing perennial energy crops is thus considered. Also [20], highlights the opportunity cost principle in farmers' decision to grow perennial energy crops and the significantly different perception of risk associated with conventional crops and perennial energy. At the same time, it emphasises the flexibility in the decision to grow conventional annual crops and thus the lower business risk compared to perennial energy crops.

The necessity of including an adequate level of risk when determining the production costs of biomass from perennial energy crops is also emphasized by Ref. [21]. For the calculation of biomass production costs, a methodology is proposed, where these costs include both the intrinsic costs associated with biomass production and the cost of land and risk. The cost of land is determined as an opportunity cost derived from crop production. Thus, a rise in grain prices leads to an increase in land cost and thus to an increase in the biomass production cost. The (annual) biomass production cost are calculated using the NPV of all costs associated with the production of biomass from perennial crops over the lifetime of the plantation. The NPV is then multiplied by the present value factor to obtain the annual cultivation cost. A discount of 6 % was used to discount and calculate the annual cultivation cost. Similarly [22], uses an approach based on the principle that farmers will primarily use an economic rationality perspective in their decision making, and thus demand the same economic benefit from farming the land regardless of the crop grown. It is proposed to use the expected price of the main crop, wheat, to calculate the biomass production price. [23] points out that the increase in the price of agricultural commodities on the global market can have a significant negative impact on the competitiveness of biomass, including biomass from perennial energy crops. Production price of biomass from perennial energy crops is derived based on the profit of conventional crops plus the (annual) production cost of biomass. Plantation establishment costs are annualized over plantation lifetime. In analysing the competitiveness of SRC biomass [24], highlights the significant impact of the increase in the price of conventional crops on the world market on the price that farmers will demand for SRC biomass.

A comprehensive comparison of the two basic approaches to determining the production costs or production price of biomass from perennial energy crops is presented in Ref. [8]. The expected biomass yields of both biomass energy crops and conventional crops are derived from soil and climatic conditions on individual land plots. A GIS-based system is then used to model the biomass production price of SRC and *Miscanthus* plantations and subsequently to determine the biomass potential according to the production price. The analysis shows the significant influence of agricultural commodity prices on the production price of biomass and its increase compared to the production cost determined using the minimum price principle to ensure an adequate return on capital employed.

The determination of the biomass production price based on the opportunity cost principle is based on finding a balance between the present value of the economic effects of growing conventional crops and the present value of biomass production from perennial energy crops. A detailed methodological approach based on modelling the cash flows associated with both types of production over the lifetime of the plantation of the perennial energy crop under consideration is presented in Ref. [8]. Biomass production price is calculated using the following formula:

$$c_{alt,1} : \sum_{k=1}^{T_{lf}} \left[c_{alt,1} \times QBio_k \times (1+i)^{(k-1)} + SBio_k - EBio_k \right] \times \left(1 + r_{n,Bio}\right)^{-k} = \sum_{k=1}^{T_{lf}} \left(Qcon_{q,k} \times Ccon_{q,k} + Scon_{q,k} - Econ_{q,k}\right) \times \left(1 + r_{n,con}\right)^{-k} \tag{1}$$

where.

$C_{alt,1}$ …. Production price of biomass from a given type of perennial energy crop in the first year of the plantation lifetime [EUR/GJ].

T_{lf} … Lifetime of plantation of perennial energy crop [years].

i … Long term average inflation [-].

$QBio_k$ … Biomass yield of a given perennial energy crop in year k on a given plot [GJ].

$SBio_k$ … Subsidy for the energy crop in year k [EUR/ha].

$EBio_k$ … All one-off and operating expenses of the energy crop plantation incurred in year k, including finance costs, taxes, etc. [EUR/ha].

$r_{n,Bio}$ … Nominal discount applied for discounting the cash flows of a given perennial energy crop plantation [−].

$Qcon_{q,k}$, …Yield of the q-th conventional crop in the recommended crop rotation on the plot in year k [t/ha].

$Ccon_{q,k}$ … Price of the q-th conventional crop in the year k [EUR/tonne].

$Scon_{q,k}$ … Subsidy per q-th crop in the year k [EUR/ha].

$Econ_{q,k}$ … One-off and operating expenditure of the q-th conventional crop incurred in the year k, including finance costs, taxes, etc. [EUR/ha].

$r_{n,con}$ … Nominal discount applied for discounting the cash flows associated with the conventional crops [−].

The discount rate values generally differ due to the different risks associated with growing annual conventional crops ($r_{n,con}$) and perennial energy crops ($r_{n,bio}$). To calculate the biomass production price of perennial energy crops, it is therefore necessary to identify and quantify the risks associated with both types of production (see below).

1.2. Determination of the discount rate for calculating the production price of biomass from perennial energy crops

Only a limited number of papers on this topic can be found in the literature, or estimates of the discount rates are published, usually without a detailed discussion of the determination of their amount. For example [18], uses a value of 6 % for discounting as an estimate of farmers' cost of capital. A similar discount rate of 6 % (without further detailed discussion) is also used by Ref. [21] in analyzing the production costs of biomass from both perennial energy crops and annual crops. A discount value of 7 % is used by Ref. [24] as a baseline for the analysis of the competitiveness of biofuels produced from cultivated biomass, pointing out both the effect of the discount level on the competitiveness of different biofuel production and use strategies and the fact that higher risk should be reflected in the discount value. In Ref. [8], the setting of the discount value is discussed in terms of the different risks associated with conventional agricultural production and biomass production from perennial energy crops. The discount value of 10 % is derived from the expected profitability of RES projects for electricity and heat production in the Czech Republic. The expected profitability of RES projects in the Czech Republic is in the range of 6–7% and the discount value of 10 % for calculating the production price of biomass for energy purposes was estimated after a premium for the higher risk of biomass cultivation business. However, even here the issue of setting an objective discount rate for discounting cash flows associated with the production of conventional crops or biomass from perennial energy crops is not systematically addressed.

Annual crops are planted and harvested within a single growing season, typically lasting from a few months to one year. The financial risk associated with growing annual crops includes the uncertainty of weather conditions, pest and disease outbreaks, and fluctuations in market prices. Since annual crops have a shorter lifecycle, the risks associated with growing them are generally considered lower than for perennial crops [19,25].

Perennial crops, on the other hand, are those that have a lifespan of 10–25 years (e. g. *Miscanthus* and fast growing trees in SRC) and produce a harvest either each year (e. g. *Miscanthus*) or cumulatively in several year rotation cycle (typically 3–6 years for SRC) [26]. Perennial crops are characterized by significant one-off establishment costs and biomass production increases gradually over the lifetime of the stand/plantation. This is particularly characteristic of SRC, with production peaking in second decade of plantation growth, typically between year 8–12 in current conditions and varieties used in commercial practice in Mild-temperate climatic zone [27]. It can be also expected that introduction of new varieties in close future of poplar, willow or other species (e.g. *Paulownia*) which have better production potential and earlier onset of yields, may improve yields and economic parameters of SRC. Another possibility to improve income and lower business risks is to optimize plantations parameters for individual soil-climate conditions and market situation including of length of rotation, spacing and density of trees. This would allow not only increase yields, but also diversification of final products with higher value-added, e.g. from woodchip to thin wood assortments for wood industry [28].

Financial risks associated with growing perennial energy crops include among other:

- **Market volatility:** The market for energy products such as biofuels can be subject to significant price volatility, influenced by factors such as government policies, changes in supply and demand, and competition from other energy sources.
- **Climate and weather risk:**
  - o During establishment of plantations (especially in the first two years): The growth and yield of perennial energy crops can be significantly affected by weather and climate conditions, such as drought and frost. Climate change is also causing greater weather volatility, which can increase the risks of financial losses due to crop failure or reduced yields [29].
  - o After establishment of plantations: Climate and weather risks of well-established plantations of perennial crops are much lower (compared with the annual crop), because they are deep-rooted (poplar willow) or rhizomatous (*Miscanthus*, Schavnat). Their good adaptation to current effects of climate change was documented in field conditions [29].
- **Production costs:** The costs of growing and maintaining perennial energy crops can be significant, including expenses for the plantation establishment, seedlings, fertilizers, pest control, harvesting cost, and land rent costs. One of the significant risks of rising costs, especially in Central and Eastern European countries, is the cost of land rent. This results from the large differences in the price of agricultural land and the level of agricultural land rents in EU member states [30].
- **Changes in subsidies:** Subsidies for conventional crops significantly increase the economic efficiency of their cultivation and thus affect the production price of biomass from energy crops. The higher the subsidies to conventional crops, the higher the farmers' expectations of the biomass production price [6]. The level of future subsidies and, in particular, the way they are allocated are subject to change, e. g. currently due to expected changes in agricultural policy in view of the implementation of the so-called Taxonomy [31].
- **Duration of land rent:** In some EU countries, the share of rented land farmed by farmers is high (e.g. 72.7 % in Czech Republic, year 2020 [32]). The typical length of the land rent can be significantly shorter than the lifetime of perennial energy crop stands/plantations (e.g. 5–6 years in the Czech Republic [32]). Thus, there is a significant risk of the landowners terminating the land lease before the end of the plantation lifetime. This risk is essentially negligible in the case of conventional agricultural crops.
- **Learning curve effect:** Biomass yields of both conventional and perennial energy crops are significantly influenced by improvements in agrotechnologies and, in particular, seed and planting materials. Annual conventional crops allow an immediate response to these developments, whereas in the case of perennial energy crops the establishment of a plantation largely fixes the yield potential for the lifetime of the plantation.

Overall, while growing perennial energy crops can offer potential returns, it is important to carefully evaluate the risks involved and develop strategies to manage these risks. Investors should consider diversifying their (business) portfolio, investing in crop insurance, and staying up to date on market trends and technology developments in the biofuels industry.

1.3. Main objectives and key novelty of the study

A number of authors stress the need to consider the higher level of risk associated with the cultivation of perennial energy crops when estimating their production price [19–21] and use various, usually qualitative analyses or expert estimates to estimate the amount of this risk. What is missing in the literature is a proposal for a systematic approach to estimating the different risks of growing conventional and perennial energy crops based on an analysis of the variability of economic effects (net profits in each year of the period under evaluation).

The main objective of this paper is to contribute to filling this scientific gap and propose a methodology to quantify the different levels of risk associated with growing conventional annual crops and perennial energy crops.

The main contribution of this study is the proposal of a general procedure for quantifying the level of risk for perennial energy crops using the Capital Asset Pricing Model (CAPM) and Monte Carlo simulation. The evaluation of the higher risk level associated with the cultivation of perennial energy crops compared to conventional annual crops allows for a correct determination of the biomass production price of perennial energy crops as a basis for farmers' decisions on which type of production to focus on. Correct determination of the business risk associated with the production of perennial energy crops is also the basis for the development of a strategy for the development of biomass use at regional or national level. The proposed procedures allow for targeted setting of government policies for the efficient development of biomass production using perennial energy crops, taking into account the specific business and financial risks involved.

The proposed methodology is generally applicable for determining the level of risk associated with biomass cultivation using perennial energy crops. The application of the proposed methodology is tested on a case study for the Czech Republic and for two perennial energy crops considered key for the climatic conditions of Central Europe (*Miscanthus* and SRC). The results of the case study confirm the often mentioned in the literature higher riskiness of growing perennial energy crops compared to annual conventional crops and the necessity to take this higher risk into account in the economic evaluation of the competitiveness of biomass from perennial energy crops.

2. Methods

2.1. Approaches to pricing biomass from perennial energy crops

The Capital Asset Pricing Model (CAPM) is a financial model used to determine the expected return on an asset, given its risk and the return on the market as a whole [33]. The model assumes that investors are risk-averse and rational and that they aim to maximize their returns while minimizing risk.

The CAPM formula uses the following inputs:

- Risk-free rate of return: The rate of return on a risk-free investment such as a high-quality governmental bond.
- Beta coefficient: A measure of the asset's sensitivity to market risk. A beta of 1 means the asset moves in line with the financial market, while a beta greater than 1 indicates higher volatility.
- Market risk premium: The expected return on the market less the risk-free rate.

Using these inputs, the CAPM formula calculates the expected return on the asset:

$$E(r) = r_f + \beta \times MRP \tag{2}$$

where E(r) … expected return [%]

r_f … risk-free rate [%]

β … beta coefficient [−]

MRP … market risk premium [%]

The beta in the CAPM depends on the time period it is calculated. The beta is a measure of the systematic risk of an asset relative to the overall (financial) market, and it is used to estimate the expected return of the asset according to the CAPM formula. In other words, beta measures the risk added by an investment into a diversified portfolio. However, investors in private companies often do not have an opportunity to diversify as most of their wealth is invested in their private businesses. In that case, the use of total beta, which includes non-systematic risk, can be more appropriate [34]. To be more exact, beta and total beta are extreme points of risk measures used in CAPM, and a non-diversified investor's risk will be closer to total beta [35].

Cenesizoglu and Reeves [36] developed three component beta model – conditional beta model. It is argued conditional beta of an asset as the component structure with short (one month), medium (one year), and long (10 years) run components. It was concluded that short-period beta captures the highest volatility while long-period beta is tied to the business cycle.

These results contradict the risk of farmers, who bear a higher risk when growing perennial energy crops. In addition, with annual crops, it is easier to react to changes in the market, demand, and unexpected changes in political decisions. Cultivation of annual crops carries a real option to change production and thus considerably reduces the financial risk of investors.

Summarizing the above, nominal discount for growing perennial energy crops and the nominal discount for growing conventional crops contain both systematic and individual risk. The theory of portfolio states that risk is proportional to the standard deviation of returns [37]. This can be used when determining the discount rate. The beta coefficient can be defined as follows:

$$\beta_j = \frac{\sigma_{jM}}{\sigma_M^2} = \frac{\rho_{jM} \times \sigma_j \times \sigma_M}{\sigma_M^2} = \rho_{jM} \times \frac{\sigma_j}{\sigma_M} \tag{3}$$

where β_j … beta coefficient for the j-th risk asset.

σ_{jM} … covariance of the returns of the j-th risk asset and the market portfolio

σ_j … standard deviation of returns of the j-th risk asset

σ_M … standard deviation of returns of the market portfolio

ρ_{jM} … correlation coefficient between returns of the j-th risky asset and the market portfolio.

In the CAPM model, the beta coefficient represents a measure of systematic risk. To include whole individual (non-systematic) risk, the correlation coefficient should be equal to one. In this case beta will be equal to so-called Total Beta. The level of investors' diversification determines the correlation coefficient used for beta calculation.

The methodology described above for determining the standard deviation of financial returns proposed for conventional crops cannot be directly used for perennial energy crops. In many countries, the cultivation of perennial energy crops is still in its starting phase, plantations are limited in size and often rather experimental. Thus, there is a lack of long-term robust data to analyse the variability of biomass yields both in tonnes of biomass and variability of revenues in financial units.

The main risk related to SRCs is the loss of the entire plantation in the first two years (namely due to drought periods or severe frosts or other weather extremes). Establishing a SRC plantation is investment intensive (establishment costs are up to 50 % of total plantation expenditures during the whole lifetime, in present value [16]) and therefore the loss is very significant for the investor. A similar risk exists in the case of non-woody perennial crops, although the share of one-off plantation establishment costs is lower. *Miscanthus*, for example, has a share of establishment cost on total cost during plantation lifetime (in present value) between 40 and 45 % [38].

A simulation model is proposed to find the standard deviation of the returns. The model is based on the estimation of probabilities p_1 and p_2, where p_1 indicates the probability of major damage to the plantation due to extreme weather (leading to the liquidation of the plantation) in the first year after establishment and p_2 the probability of major damage to the plantation in the second year after establishment. The probability of a perennial energy crop plantation being fatally damaged in subsequent years is then very low and can be ignored [38,39]. Beyond these probabilities, other model inputs are:

- the ratio of plantation establishment expenditures to average year CF respectively (hereinafter referred to as loss ratio v_r, typically it

represents number of average annual net cash flow losses, undiscounted),
- plantation lifetime,
- variability of cash flows due to variability of biomass yields (assuming standard weather conditions),
- probability of yield loss in one year due to bad weather conditions and the ratio of the cost of lost harvest to the cash flow of a standard year.

The standard deviation of cash flows (measured relative to the mean of year cash flows) is then computed via simulation Monte Carlo. For the simulations of the standard deviation for SRC and *Miscanthus*, values of plantation parameters typical for Czech conditions were used.

Fig. 1 shows the dependence of the magnitude of the standard deviation of returns of SRCs plantations on the loss probability p_1 and loss ratio v_r. The probability p_2 is assumed to be zero. The lifetime of the plantation is 25 years and the variability of biomass yield is 10 %. The simulation assumes that biomass yields in a three-year rotation are not significantly affected by adverse weather in one year of the rotation. The three-year harvest period was chosen according to the typical time between harvests in the Czech Republic [29].

Fig. 1 shows that the resulting volatility of cash flows for SRC grows approximately one to one to the probability of losing the plantation in the first year as well as in relation to the loss ratio v_r. For the values known from experimental plantations in the Czech Republic ($p_1 = 0.075$ and $v_r = 7$), the value of the standard deviation is approximately 10 %.

The simulation results of the standard deviation of cash flow for *Miscanthus* are shown in Fig. 2. The simulation assumes following input data: loss probability p_1 is 5 % (p_2 is half of p_1), plantation lifetime is 15 years, variability of annual cash flow is 15 %, loss ratio v_r is 5 and probability of loss due to bad weather condition in one year is 5 % (any year after first two years after plantation establishment).

The resulting standard deviation for *Miscanthus* is at most twice that for SRC.

Since the total standard deviation is needed, it is necessary to include deviations in annual cash flows in the calculations (resulting from variability of annual yields due to the weather conditions), i.e. in the models mentioned above, the relative annual cash flow is used (expected annual cash flow without influence of adverse weather conditions divided with modelled cash flow in a given year). A uniform distribution was used in the simulation.

Figs. 1 and 2 show the results of simulations where only parameters p_1, p_2 and v_r were considered. Similar simulations were carried out with the inclusion of additional parameters incorporating variations in cash flows due to variations in crop yields The objective was to obtain a formula for determining the overall variability in cash flows without having to run the simulations repeatedly. By linear interpolation of the results of the simulations, formulas were derived for SRC and *Miscathus* that allow direct estimation of the total cash flow variability given knowledge of the variability in biomass yields of a given crop and the variability associated with the risk of plantation damage after establishment (obtained from the simulations):

$$\sigma_{SRC} = \sigma_{SRC\_h} + 0.9 \times \sigma_{SRC\_CF} \tag{4}$$

$$\sigma_{Mis} = \sigma_{Mis\_h} + 0.8 \times \sigma_{Mis\_CF} \tag{5}$$

where.

σ_{SRC}, σ_{Mis} … the whole risk standard deviation of CF for SRC or *Miscantus* growing

σ_{SRC_h}, σ_{Mis_h} … standard deviation of CF for SRC or *Miscantus* related with the risk of bad weather leading to plantation damage

σ_{SRC_CF}, σ_{Mis_CF} … the standard CF standard deviation of CF for SRC or *Miscantus* due to the natural variability of biomass yields.

3. A case study of the application of the methodology for the Czech republic

Using the Czech Republic as an example, the next step is to

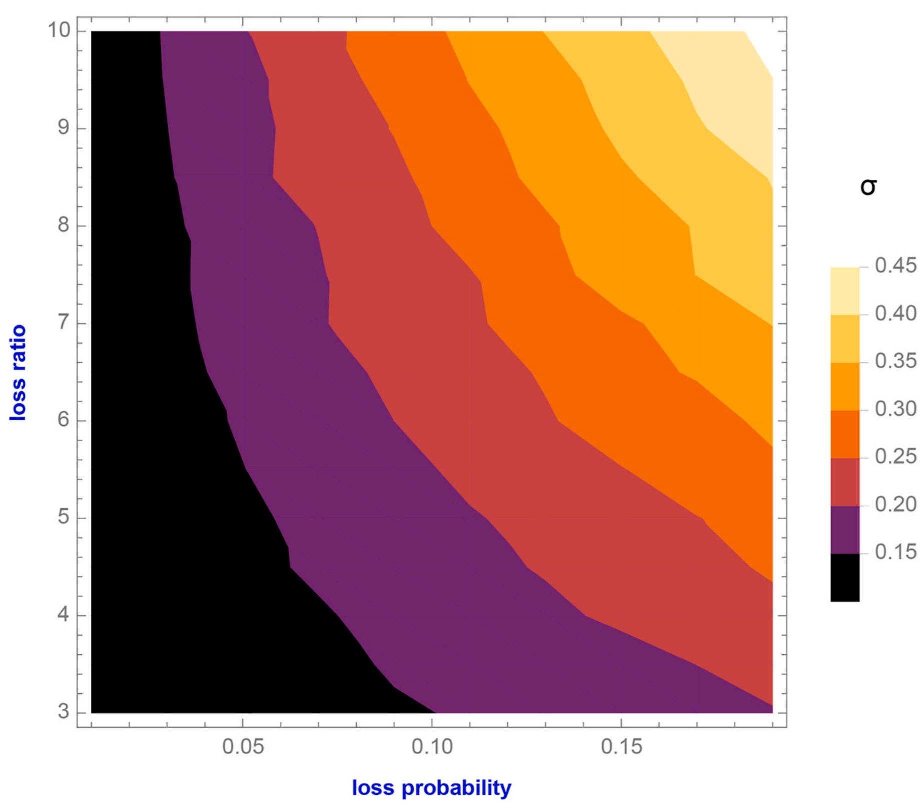

Fig. 1. Demonstration of simulation results of standard deviation of cash flows for SRC.

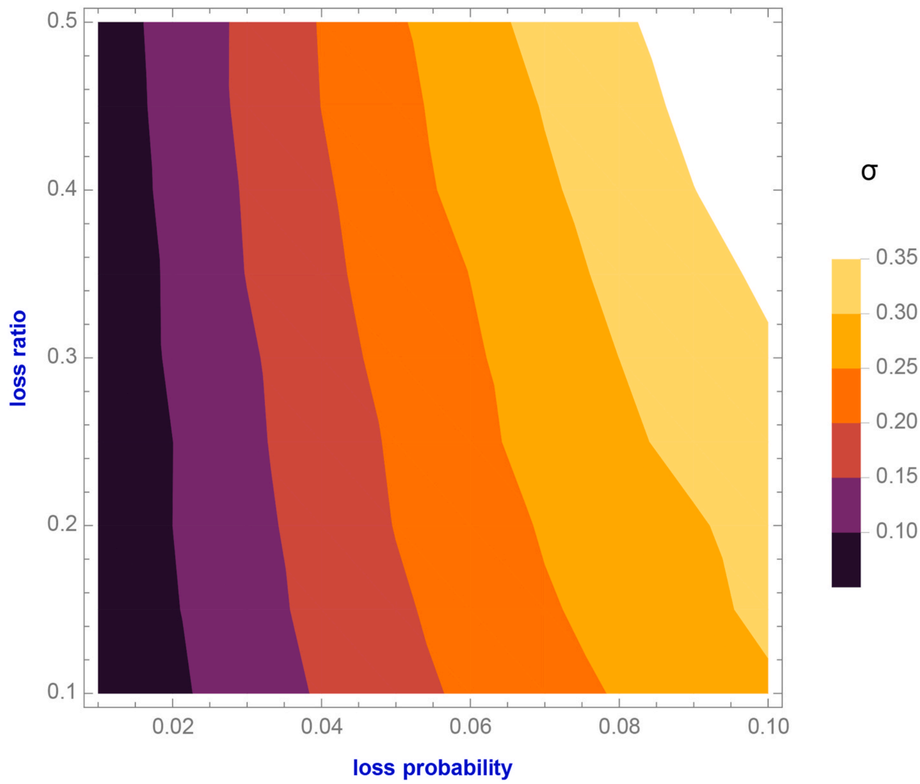

Fig. 2. Demonstration of simulation results of standard deviation of cash flows for *Miscanthus*.

demonstrate the application of the methodology for determining the discount rate for conventional and perennial energy crops. At the same time, this chapter demonstrates the effect of a higher discount for perennial energy crops on the biomass production price.

Data on conventional crop yields, conventional crop prices, and subsidies and cultivation costs for the period 2010–2021 are used for the analysis of yields and cash flow variability. Data on yields and cultivation costs of perennial energy crops are mainly based on experimental plots and analysis of the processes required to implement and operate SRC and *Miscanthus* plantations [8,38].

The period since 2010 can be considered as the period when the transformation of agriculture started since 1990 was completed. Between 1990 and 2010 there were changes in land ownership, farming methods, availability of subsidies, etc. This has had a significant impact on crop yields, the structure of the crops grown and, not least, on cultivation costs [40].

To demonstrate the application of the methodology, the three most important conventional crops with the highest share of the 2021 cropped area were used, namely wheat (32 %), barley (13.3 %) and rape (14 %) [41].

3.1. Data for biomass production price modelling

The production price of biomass from perennial energy crops was modelled using Eq. (1). Data on individual arable land parcels characterized by a unique combination of soil and climate properties were used for modelling (database Land Parcel Identification System – LPIS). The model uses a total of approximately 570 allowable combinations of climate region and soil type from which to derive long-term expected yields for both conventional and energy crops. The combinations of climate region and soil type define the so-called 'main soil unit' (MSCU) [42].

The three most important agricultural production areas - potato, beet and maize - were used for the biomass production price analysis. Unique MSCU codes are assigned to each area [6]. These areas cover the dominant part of the seeded area in the country (maize area - 6.2 %, beet area - 45.2 %, potato area - 41.7 %) and are characterized with the similar planting cost [41].

On individual plots, it is assumed that conventional crops will be rotated according to recommended sowing practices reflecting the soil and climatic conditions of individual agricultural production areas in the Czech Republic. Crop rotation is assumed in a seven-year cycle [43].

All arable land registered in LPIS was included in the modelling. However, the selection of land for analysis can be restricted in the model according to soil and climate characteristics so that land with the highest yields of conventional crops is not included (and therefore not considered for biomass production from perennial energy crops).

3.2. Cultivation cost of conventional crops

Crop cultivation costs are taken from a database of agricultural norms that tracks the cost of individual crops in agricultural production areas [44].

For the calculation of the effect of the change in the discount rate on the biomass production price, the crop cultivation cost for the year 2021 (latest available values) were used as the initial production cost - see Table 1.

Changes in the cultivation cost of crops over the period 2010–2021 are demonstrated in Fig. 3 for the beet-growing area.

Table 1
Range of production costs of three basic conventional crops, nominal values, Czech Republic, 2021 [44].

Cost range (EUR/ha, year)		
Wheat 1118–1246	Rape seed 1414–1558	Barley 979–1057

Note: Exchange rate 24 CZK/EUR.

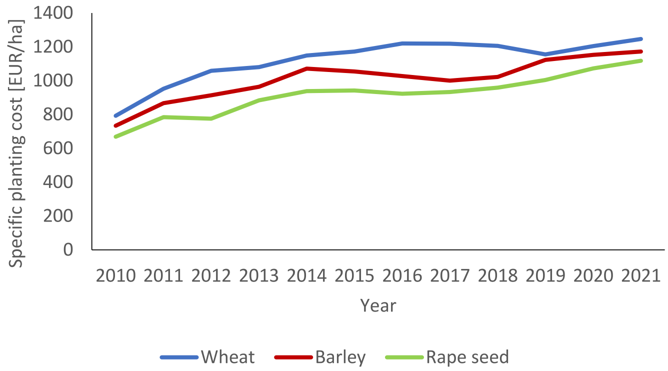

Fig. 3. Development of cultivation costs for the beet growing area, nominal values, Czech Republic [44].

3.3. Data on direct subsidies for conventional crops

Subsidies represent the total subsidies per hectare in a given year. These are the subsidies provided by the EU (SAPS - single area payment scheme) and the national top-up (Top-Up or Greening). The total subsidy per hectare in 2021 was EUR 207. The initial subsidy per hectare in 2010 was EUR 187, peaking in 2013 at EUR 243 [45].

3.4. Agricultural commodity prices data

Agricultural commodity prices were taken from the (monthly) reports for cereals, oilseeds and fodder, with annual values calculated from monthly averages [46]. The evolution of the prices of the three commodities used for the discount calculation is shown in Fig. 4. For the modelling of the production price of biomass from energy crops, the average prices of 2021 were then used.

3.5. Yields of conventional and energy crops

Data from crop yield statistics for individual regions of the Czech Republic were used to analyse the variability of yields of conventional crops. The total area of sown land in the Czech Republic in 2021 was 2452 thousands ha. The Czech Republic is divided into 14 regions (NUTS3 in EU nomenclature). The crop yields are used for the period 2010–2021 by region and their variability from year to year for wheat is shown in Fig. 5.

Data on *Miscanthus* (*Miscanthus* × *giganteus*) yields are taken from field trials on experimental plantations of the Silva Tarouca Research Institute (VUKOZ) in different sites of the Czech Republic [26]. *Miscanthus* yield variability was estimated to be 15 % based on data from individual sites.

Miscanthus is sensitive to winter frosts after planting, which can significantly damage a newly established plantation. Sensitivity to winter frost varies between genotypes or clones and the critical temperature of rhizomes is given as between −0.4 and −5.9 °C depending on the genotype [47]. The number of surviving individuals after the first year from the establishment of experimental plantations ranged from 84 to 94 % [26]. Conservatively for the conditions of the Czech Republic, the probability of fatal damage to a *Miscanthus* plantation in the first year after establishment can be estimated as about 5 %, and in the second year after establishment as 1–2.5 %. In subsequent years it is then negligible and can be considered to be zero.

Variability of SRC yields and the probability of fatal damage to the newly established plantation were also derived from the results of field trials on experimental plantations of VÚKOZ. The variability of SRC yields (due to the three-year rotation period) is typically lower than in the case of *Miscanthus* and can be estimated to be about 10 % for similar site conditions. This is due to the fact that poorer conditions in one year can be compensated in the following years of the cropping rotation. From experimental and trial plantations, the risk of serious damage to the plantation leading to the need to re-establish the plantation (or abandon the project) can be estimated at approximately 7.5–10 %.

To model the yield variability of both perennial crops, the following inputs were used as inputs for the Monte Carlo simulations.

3.6. *Miscanthus*

- Lifetime of the plantation: 15 years
- Probability of plantation damage in the first year after establishment is 5 %, and probability of the damage in second year is 2.5 %, investor depreciates the investment in establishment and loses on average about 5 annual net cash flows in the sum (undiscounted)
- Variability of annual cash flows: 15 % due to biomass yields fluctuation
- Probability of loss of harvest in a given year (without damage of the plantation) is 5 %, in a given year the cash flow consists only of expenses equal to 40 % of the usual annual cash flow

3.7. SRC

- Plantation lifetime: 25 years
- Probability of plantation damage in the first year after establishment is 7.5 %, the investor depreciates the investment in the establishment and loses on average about 7 annual net cash flows in total (undiscounted)

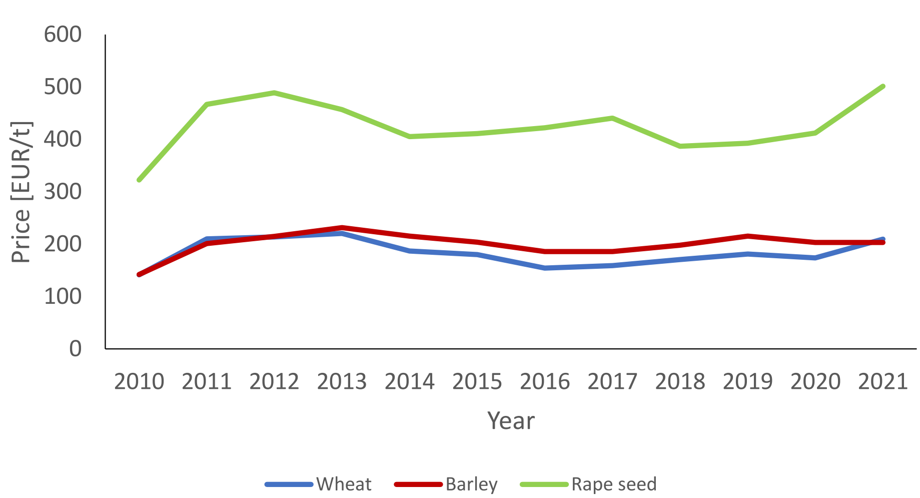

Fig. 4. Wheat, rapeseed and barley prices in nominal values, the Czech Republic [46].

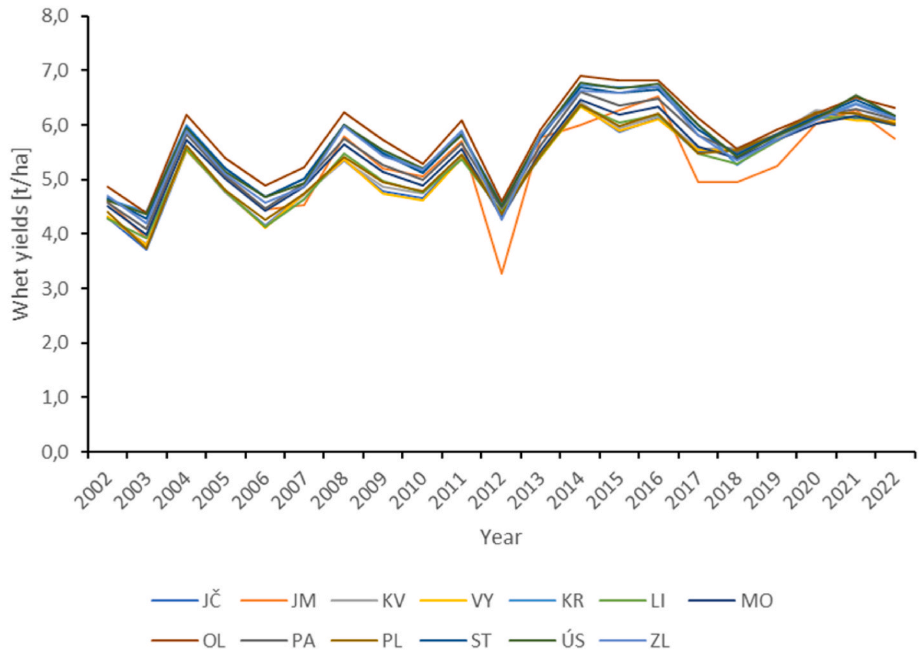

Fig. 5. Variability of wheat yields in individual regions of the Czech Republic (except the capital city of Prague) [41].

- Variability of annual cash flows: 10 %

3.8. Financial market return data for determining the discount value

The input data on market returns were the market and sectors return according to Ref. [48]. The expected volatility of the Czech market return is 20 % [49]. It is similar to the volatility of markets from other European countries. The market risk premium for the Czech Republic is 6.97 % [50] and the risk-free rate is currently 4.38 % based on the average of 10-year Czech government bond yields for 2022–2023 [51].

4. Results

The average volatility of returns (cash flows) for the three analyzed conventional crops (wheat, barley, rape seed) is 12 % based on the data from the period 2010–2021 and for each region of the Czech Republic. Based on the input data and Monte Carlo simulation results, the volatility for *Miscanthus* and SRC is 21.1 % and 19.7 % respectively. Due to the relatively slight difference, an average value of 20.4 % can be used.

The risk premium, exceeding the risk-free rate, should reflect volatilities proportionally. Beta for both annual crops and perennials is calculated by Eq. (3). Since an individual farmer cannot be considered a diversified investor, the correlation is set to one in that case.

Market risk premium for the Czech Republic 6.97 % and the risk-free rate 4.38 % lead to the discount rate **8.5 %** for annual crops with a beta equal to 0.34 and the discount rate for perennials at **11.5 %** with a beta equal to 0.6. This is the average value for *Miscanthus* and SRC.

Calculated discount rates are affected by high risk-free interest rates, which have risen enormously in period 2022–2023 (4.38 % versus 1.82 % as the long-term average of the risk-free rate over 10 years). Based on this fact one can state that calculated discount rates (both for conventional and energy crops) are higher than before the start of quick inflation growth since the end of 2021.

The production price of biomass from SRC and *Miscanthus* plantations was analyzed against three discount rate scenarios for energy crops: S1 - the same discount rate as for conventional crops is used for energy crops - i.e. 8.5 %, S2 - the discount rate determined by the CAPM model is used for energy crops - 11.5 %, S3 - the discount rate increased by 3 percentage points for energy crops compared to the S2 scenario (e.g. reflecting risks not included in the analysis). At the same time, a long-term average inflation assumption of 2 % was used for both crop types. This value was also used to index cultivation cost and crop/biomass prices both conventional crop prices and biomass from perennial plantations.

The summary results of the modelling of biomass production price from SRC and *Miscanthus* plantations are presented in the following Table 2 and Table 3.

5. Discussion

The production price of biomass from SRC plantations is significantly more sensitive to the increase in the discount rate value for energy crops relative to conventional crops than is the case for biomass from *Miscanthus* plantations. The primary reason for the higher sensitivity of SRC plantations to a change in the discount rate is the higher share of one-off establishment costs in the total cost (in present value) of the plantation over its lifetime. At the same time, SRC plantations do not reach peak production until between the eighth and twelfth year after establishment. This further reduces the present value of future revenues as the discount value increases.

The variability of the biomass production price from SRC plantations for the maize and potato agricultural production areas is presented in Figs. 6 and 7. For both selected agricultural production areas, the results of biomass production price modelling for three scenarios of discount rate for energy crops are presented here by combining the discount values for conventional and energy crops.

Figs. 8 and 9 present analogous figures on the variability of biomass production price from *Miscanthus* plantations for scenarios S1, S2 and S3.

Table 2
Production price of biomass from SRC plantations and impact of different scenarios for discount rate used for energy crop, Czech Republic data, 2021.

	S1	S2	S3	S2/S1	S3/S1
	[EUR/GJ]	[EUR/GJ]	[EUR/GJ]	[–]	[–]
Maize region	9.5	11.8	16.1	24.2 %	69.5 %
Sugar beet region	7.1	8.8	11.9	23.9 %	68.3 %
Potato region	6.7	8.3	11.2	23.9 %	66.7 %

Table 3
Production price of biomass from *Miscanthus* plantations and impact of different scenarios for discount rate used for energy crop, Czech Republic data, 2021.

	S1	S2	S3	S2/S1	S3/S1
	[EUR/GJ]	[EUR/GJ]	[EUR/GJ]	[–]	[–]
Maize region	7.4	8.7	10.1	16.5 %	35.2 %
Sugar beet region	6.4	7.4	8.2	15.6 %	28.1 %
Potato region	10.8	12.6	13.9	16.7 %	28.7 %

Note: The biomass production price is set as a weighted average value for a given agricultural production area, where the weight is the area of each plot.

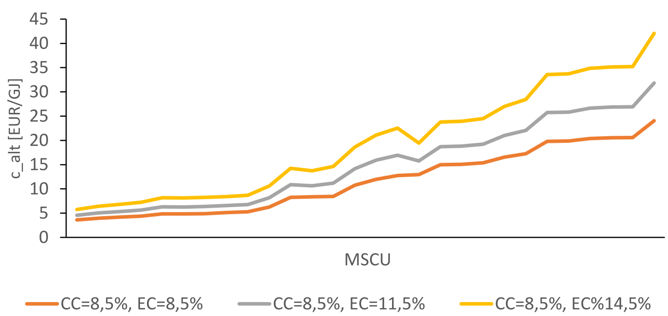

Fig. 6. Variability of biomass production price from SRC plantations, maize production, region, Czech Republic, 2021 data.

The modelling results show a significant effect of the higher risk associated with growing perennial energy crops on the expected biomass price (i.e. biomass production price) by biomass growers. Some risks that are connected to cultivation of perennial energy crops are not included, such as land lease periods that are shorter than the lifetime of the plantations and the volatility of future biomass demand. Rational decision-making by investors can be assumed, and the determination to grow perennial crops can be expected to pertain only to farmers who owned their land or who hold a long-term lease. Similarly, the risk associated with future demand for biomass can be considered rather negligible, given the general trend of replacing conventional fossil fuels with renewable energy sources.

On the other hand, the risks associated with changes in farmland management conditions (reduction in the use of artificial fertilisers and chemicals) leading to a possible decrease in hectare yields are significant potential future risks. This could lead to an increase in agricultural commodity prices and consequently to an increase in the production price of biomass. A significant impact on the biomass production price would come from a change in the level of agricultural subsidies, especially in the case of linking subsidies to sustainable use of agricultural land and provision of ecosystem services. This in turn could positively affect (reduce) the biomass production price because part of the income from plantations of perennial energy crops would come from payments (subsidies) for ecosystem services [7].

In 2022, there are significant increases in both agricultural input prices and crop prices. For example, the increase in agricultural input prices between 2021 and 2022 was 24 % [52]. Conversely, the increase in commodity prices was even higher, e.g., wheat prices increased by 68 % between 2021 and 2022, rapeseed by 42 %, and barley by 45 % [46].

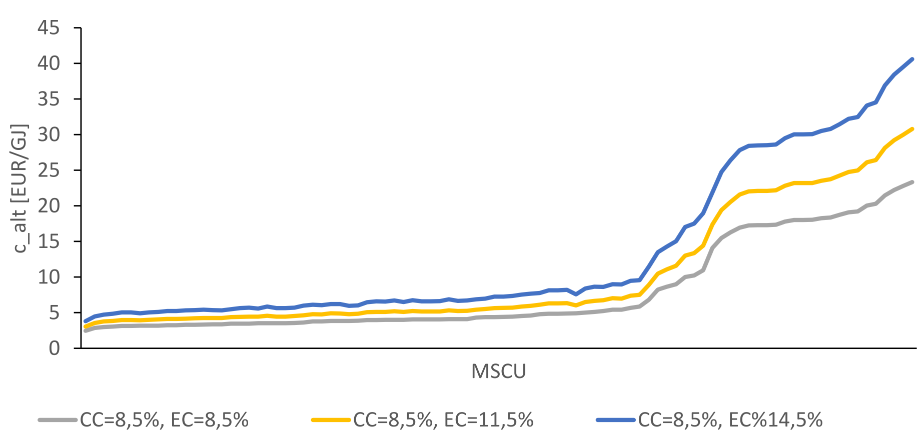

Fig. 7. Variability of biomass production price from SRC plantations, potato production region, Czech Republic, 2021 data.
Note: The values of the main soil unit on the horizontal axes are ordered according to the increasing value of the biomass production price for scenario S1. This is to increase the clarity of the figures.

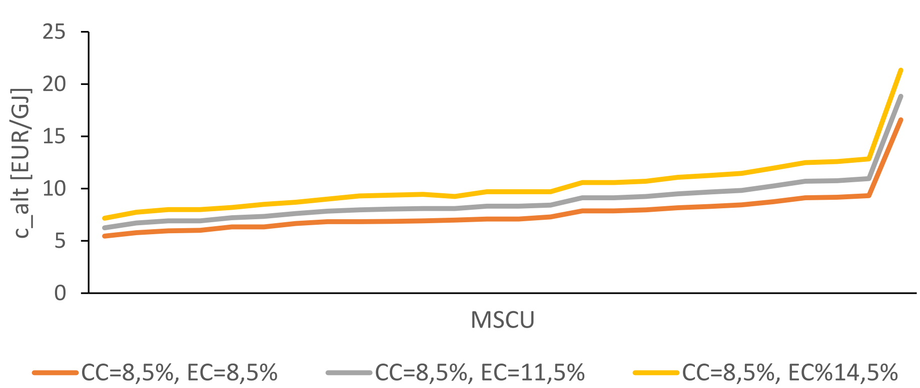

Fig. 8. Variability of biomass production price from *Miscanthus* plantations, maize production region, Czech Republic, 2021 data.

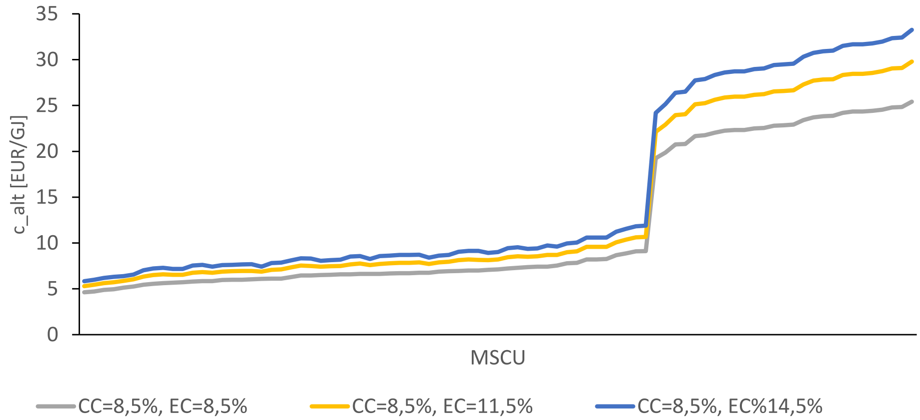

Fig. 9. Variability of biomass production price from *Miscanthus* plantations, potato production region, Czech Republic, 2021 data.

Although agricultural commodity prices decline in 2023, they remain at significantly higher levels compared to 2021. If the biomass production price modelling were performed at the 2022 price level, this would lead to, for example, an increase in the SRC plantation biomass production price for the maize production area from 11.8 EUR/GJ to 13.1 EUR/GJ and for the potato production area from 8.3 EUR/GJ to 9.5 EUR/GJ.

An important aspect in setting the discount rate for perennial energy crops is also the variability of their yields and the probability of plantation damage after establishment. These values can be reduced (and thus reduce the risk associated with the cultivation of perennial energy crops) by appropriate selection of planting material and plantation sites. Thus, the risk associated with the cultivation of perennial energy crops can be expected to be stable with a slight downward trend in the future.

The risks associated with establishing plantations of perennial energy crops can be significantly reduced by changing the subsidy policy in the form of shifting at least part of the operating (annual) subsidies to subsidies for establishing plantations of perennial energy crops. Risks can be further reduced by providing advice on the establishment of perennial energy crop plantations for the selection of suitable sites and the appropriate type of planting material.

6. Conclusions

Perennial energy crops compete with conventional annual crops in land-use. The production price of biomass from perennial energy crops is thus derived on the basis of the opportunity cost principle. This means that in deciding whether to grow perennial energy crops, the farmer will demand a price for biomass that will at least provide him with the same economic effect over the lifetime of the plantation in the form of the present value of net cash flows over the lifetime of the energy crop plantation. The discount rates used to discount the cash flows associated with (annual) conventional crops and perennial energy crops play an important role in modelling the biomass production price. The business risks associated with the two types of crops are different, with generally higher risk associated with perennial energy crops. The CAPM model can be used to derive the amount of this risk for both conventional and perennial energy crops. The results of the analyses carried out for the Czech Republic (at the 2021 price level) show a significantly higher risk and discount value for perennial energy crops (by about 3 %), which increase the price that producers expect on the market. The proposed methodology makes it possible to determine the discount rate for perennial energy crops while respecting the most important risks associated with plantation establishment and biomass production. The application of the methodology makes it possible to objectify farmers' decisions on crop choice. At the same time, it provides a tool for decision-making at the macro policy level, e.g. when developing State or regional energy concepts, taking into account the realistic biomass potential of perennial energy crops.

CRediT authorship contribution statement

J. Knápek: Methodology, Formal analysis, Writing – original draft. **O. Starý:** Conceptualization, Methodology. **K. Vávrová:** Investigation, Project administration, Supervision. **J. Bemš:** Methodology, Data curation, Writing – review & editing. **J. Weger:** Methodology, Data curation. **M. Horák:** Software, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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